

Mathematics, 2018 EC, entrance exam, version 4

Grade 12 · Mathematics · 2018 E.C.

Time allowed: 120 minutes

Questions: 56

Total marks: 56

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following statement is true about the set of integers? (1)
 - A. Integers only include positive numbers and Zero.
 - B. Every integer is rational number
 - C. The set of integers included irrational numbers
 - D. Every integer is also a whole number
2. Which of the following number is an irrational number? (1)
 - A. π
 - B. -7
 - C. $0.222\ldots$
 - D. $\sqrt{25}$
3. A gardener wants to build a square flower bed with an area of 45 square feet. Between which two whole numbers on a tape measure will side length fall? (1)
 - A. 8 and 9
 - B. 5 and 6
 - C. 6 and 7
 - D. 7 and 8
4. Which of the following square roots represent a rational number? (1)
 - A. $\sqrt{0.25}$
 - B. $\sqrt{10}$
 - C. $\sqrt{0.1}$
 - D. $\sqrt{2}$
5. A student solves a system of linear equations in two variables and finds $x = 4$. Then he concludes that the solution to the system is 4. Which evaluation is correct? (1)
 - A. The conclusion is correct only if the system has no solution
 - B. The conclusion is correct because x is independent variable
 - C. The conclusion is correct if the student used substitution method.
 - D. The conclusion is incorrect because the solution must be in ordered pair form.
6. What is the solution set for $|x^2 - 5| = 4$? (1)
 - A. $\{-3, -1, 1, 3\}$
 - B. $\{-3, -2, 1, 3\}$
 - C. $\{-1, 1\}$
 - D. $\{-1, 1\}$
7. A toy rocket is launched. The path is modeled by $h(t) = -16t^2 + 64t$. At what time does it reach maximum height? (1)
 - A. 2 seconds
 - B. 4 seconds
 - C. 10 seconds
 - D. 64 seconds
8. In survey of 100 students, 60 like mathematics, 50 like English and 30 like both. How many like neither? (1)
 - A. 10
 - B. 30
 - C. 20
 - D. 40

9. If set A is proper subset of set B, then their intersection is equal to: (1)
- A. $\emptyset$ B. A
C. U D. B
-
10. If a relation R satisfies: whenever $(a,b) \in R$ and $(b,a) \in R$, then the relation is said to be: (1)
- A. Transitive B. symmetric
C. reflexive D. asymmetric
-
11. A taxi service charges a flat fee of Birr 25 and Birr 20 per kilometer. How much does a 10 km ride cost? (1)
- A. 45 birr B. 270 birr
C. 55 birr D. 225 birr
-
12. Based on a graph of a linear function, what does the y-intercept represent? (1)
- A. The value of x when $y=0$
B. The point where the line crosses the vertical axis ($x=0$)
C. The steepness of the line
D. The total length of the line segment
-
13. If a relation $R = \{(x,y): y \leq x \text{ and } x \leq 4\}$, then which is true? (1)
- A. Domain = $\{x: x < 4\}$ B. Range = $\{y: y \leq 4\}$
C. Range = $\{y: y \geq 4\}$ D. Domain = $\{x: x \geq 4\}$
-
14. Which of the following sets of ordered pairs is a function? (1)
- A. $\{(1,2),(3,4),(3,-1),(-1,2)\}$ B. $\{(0,2),(6,3),(7,1),(0,-2)\}$
C. $\{(1,3),(4,6),(2,3),(-2,6)\}$ D. $\{(1,3),(2,6),(2,7),(-2,3)\}$
-
15. If the expression (x^2-5x+6) represents the area of a rectangle, what is the perimeter when $x=6$? (1)
- A. 12 B. 16
C. 18 D. 14
-
16. Which of the following is the vertex of $y = x^2 + 4x + 11$? (1)
- A. $(-2,7)$ B. $(-3,4)$
C. $(3,-4)$ D. $(2,-7)$
-
17. If $f(x)=x^3+2x^2+x-5$ and $g(x)=-2x^2+5x-3$, then degree and constant term of $f(x) \cdot g(x)$ respectively? (1)
- A. 6 and -15 B. 5 and 15
C. 2 and -8 D. 3 and 9
-
18. If $f(x)=x^{16}+8x^3+77x+99$ and $d(x)=x+1$, remainder when $f(x)$ divided by $d(x)$ is: (1)
- A. 31 B. 99
C. 15 D. 108
-
19. Which of the following is a factor of $f(x)=x^3+4x^2+x-6$? (1)
- A. $x-2$ B. $x-3$
C. $x+1$ D. $x+2$

-
31. The simplest form of $((2x^2+10x+12)/(x^2-9)) \div ((x+2)/(x-3))$, $x \neq \pm 3$ is: (1)
- A. 1
B. 2
C. 3
D. 4
-
32. Solution set of $(2x^2 - x - 21)/(2x-7)=0$, $x \neq 7/2$ is: (1)
- A. $\{-3\}$
B. $\{3\}$
C. $\{-4\}$
D. $\{2\}$
-
33. Which one of the following is not a rational function? (1)
- A. $f(x)=(3x+2)/\sqrt{x^2}$
B. $g(x)=2x^2+6x-7$
C. $n(x)=x^3/(2x+1)$
D. $k(x)=(x^4+1)/(x+3)$
-
34. Which is true about $f(x)=(3x^2+1)/(x^2+4x+4)$? (1)
- A. $x=2$ is vertical asymptote
B. $f(x)$ has a hole at $x=-2$
C. $y=0$ is horizontal asymptote
D. The line $y=3$ is horizontal asymptote
-
35. The perimeter of a triangle is 30 cm. The shortest side is 4 cm shorter than the longest side. The longest side is 6 cm less than the sum of the other two. Find the sides. (1)
- A. 6,7,8
B. 10,11,12
C. 8,10,12
D. 9,13,14
-
36. What is the area of the triangle with vertices A(1,1), B(1,3), C(4,5)? (1)
- A. 4
B. 3
C. 2
D. 6
-
37. Let A and B be 3×3 matrices with $\det(A)=2$, $\det(B)=3$. Then $\det(3AB^T)$ is: (1)
- A. 160
B. 48
C. 54
D. 162
-
38. Let A and B be 2×2 matrices with $\det(A)=2$, $\det(B)=16$. Then $\det(2A^{-1}B)$ is: (1)
- A. 32
B. 64
C. 128
D. 108
-
39. An amphitheater has 30 seats in first row, 35 in second, 40 in third, etc. If there are 30 rows, how many seats in 30th row? (1)
- A. 1775
B. 3075
C. 175
D. 275
-
40. If sum of first three consecutive terms of an AP (all positive) is 15 and sum of their squares is 83, then sum of first five terms? (1)
- A. 35
B. 45
C. 47
D. 53
-
41. What is eighth term of an arithmetic sequence whose third term is -4 and common difference is 5? (1)
- A. 21
B. 15
C. 19
D. 18

42. Consider geometric series $1 + 2/3 + 4/9 + 8/27 + \dots$ What is sum as $n \rightarrow \infty$? (1)

- A. 2
C. 5
- B. 1.5
D. 3

43. What is the value of x such that terms $(3+x)$, $(11+x)$, $(18+x)$ form a geometric progression? (1)

- A. 22
B. -67
C. 67
D. 54

44. Which condition must be met for infinite geometric series $a + ar + ar^2 + \dots$ to converge? (1)

- A. $|r| > 0$ B. $r > 0$
C. $r = 1$ D. $|r| < 1$

45. A man saves Birr 160 first month and increases by Birr 20 each subsequent month. Which sequence models total savings? (1)

- A. Harmonic B. Geometric
C. Arithmetic D. Fibonacci

46. What does the derivative of a function represent? (1)

- A. Average values
B. Value of y-intercepts
C. Total area under the curve
D. Slope of the tangent line

47. Find derivative of $f(x)=5x^3-2x+7$ (1)

- A. $5x^2-2$ B. $15x^2-2$
C. $15x^2-2x$ D. $15x^3-2$

48. Find equation of normal line to $f(x)=x^2$ at $(1,1)$. (1)

- A. $y-1 = -2(x-1)$ B. $y-1 = -\frac{1}{2}(x-1)$
C. $y=x$ D. $y-1 = -2(x-1)$

49. If graph of $f'(x)$ is a horizontal line above x-axis, what about $f(x)$? (1)

- A. Constant function
B. Increasing at increasing rate
C. Parabola opening upwards
D. Linear function with positive slope

50. Slope of tangent to $f(x)=x^2-5x+6$ at $x=3$? (1)

- A. 4
C. 1
- B. -2
D. 6

51. Find area bounded by $f(x)=x^2$, x -axis, $x=0$ and $x=3$. (1)

- A. 6
B. 9
C. 3
D. 27

52. A company's marginal revenue $MR(x) = -2x + 100$. Change in total revenue when production increases from 10 to 20 units? (1)

- A. 1000
B. 700
C. 800
D. 600

53. Find the range of the grouped data: $x:10-14,15-19,20-24,25-29,30-34$; $f:2,5,8,4,1$ (1)

- | | |
|--------------|--------------|
| A. 20 | B. 24 |
| C. 25 | D. 15 |
-

54. Find the mode of the grouped data (to two decimal places). (1)

- | | |
|-----------------|-----------------|
| A. 23.66 | B. 20.91 |
| C. 21.64 | D. 24.34 |
-

55. Find the median of the grouped data. (1)

- | | |
|------------------|------------------|
| A. 18.685 | B. 19.875 |
| C. 20.865 | D. 21.375 |
-

56. Find the mean of the grouped data. (1)

- | | |
|-----------------|-----------------|
| A. 21.25 | B. 19.45 |
| C. 18.15 | D. 16.95 |
-

— END OF EXAMINATION —